



Happy • Sad • Angry • Fear

KidMood

Classroom Emotion Recognition with FER-2013

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Solo Project

Tier 2 - Transfer Learning

ITAI 1378: Computer Vision and AI

Practical proposal focused on a realistic build, not theory.

What real-world problem are you solving?

Teachers and caregivers cannot continuously monitor every student's emotional state during class or group activities.

Why it matters

Who cares: teachers, parents, counselors, and school staff.

Missed signs of distress or anger can hurt engagement, behavior, and classroom safety.

- A lightweight emotion-recognition tool can provide an early warning signal without needing a complex system.

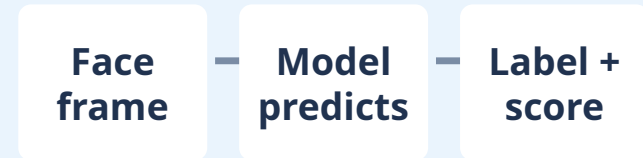
The system takes a face image or video frame as input.

A trained model predicts one of four emotions: Happy, Sad, Angry, or Fear.

- Results can be shown as a per-image label or as a simple classroom mood summary over time.

One-sentence description

KidMood is a transfer-learning system that classifies facial expressions into four emotions to support classroom awareness and engagement monitoring.



Simple, feasible pipeline

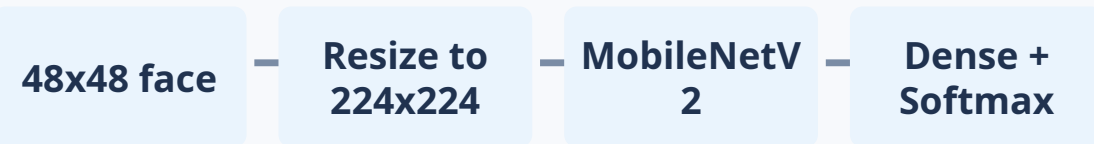
Technique: image classification for facial expression recognition.

Model: MobileNetV2 with transfer learning; backup option: ResNet50.

Framework: TensorFlow/Keras in a Colab notebook.

- Why this choice: transfer learning is faster to train, performs well on limited time/resources, and is realistic for a solo student project.

Model architecture (planned)



Frozen pretrained backbone for feature extraction

Global average pooling + small dense layer

- Final 4-class softmax output

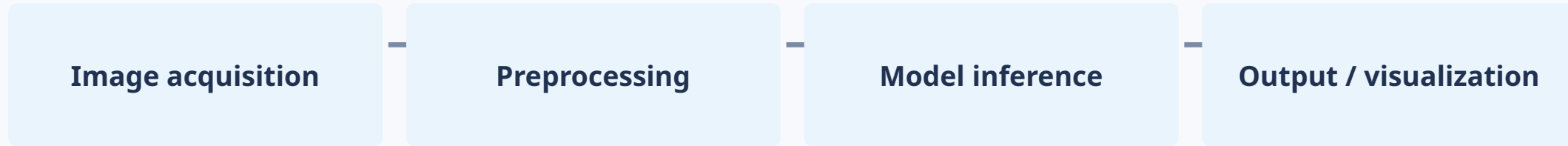
Source	Size	Labels	Preparation
FER-2013 public dataset	35,000+ face images (approx.)	Happy Sad Angry Fear	Normalize pixels Resize for model Convert grayscale to 3-channel Use light augmentation

This plan is feasible because the dataset is public, already labeled, and small enough to train with transfer learning in Colab.

System Diagram

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[Input] → [Processing Step 1] → [Processing Step 2] → [Output]



Camera or dataset image → face crop / resize → MobileNetV2 predicts emotion → show label, confidence score, or class summary.

Metric Type	Target
Primary: Accuracy	$\geq 70\%$ as a realistic first goal
Secondary: Macro F1	Balanced class performance
Speed / Latency	< 1 second per image in demo

Why use a confusion matrix?

It shows which emotions get confused with each other, especially Fear vs. Sad or Angry vs. Fear.

Sample confusion matrix template

		Predicted			
		Happy	Sad	Angry	Fear
Actual	Happy	✓			
	Sad		✓		
	Angry			✓	
	Fear				✓

Week-by-Week Plan

Slide 8

Week	Task	Milestone
10 (Oct 30)	Get dataset, set up environment	Dataset ready
11 (Nov 6)	Train / fine-tune MobileNetV2	Model working
12 (Nov 13)	Test and improve	Good accuracy
13 (Nov 20)	Create demo visuals	Demo ready
14 (Nov 27)	Final testing and documentation	Everything done
15 (Dec 4)	Present project	Presentation day

This timeline keeps the scope practical and achievable for a solo student project.

Challenge	Probability	Plan B / Mitigation
Accuracy too low	Medium	Use augmentation, tune learning rate, or try ResNet50
Classes are imbalanced	Medium	Use class weights and macro F1-score
Fear gets confused with Sad	Medium	Inspect confusion matrix and adjust training
Demo setup is unstable	Low	Use prerecorded examples and screenshots

Practical backup plans keep the project feasible.

Resource	Options / Notes
Compute	Google Colab (free) or local machine
Frameworks	TensorFlow/Keras, NumPy, scikit-learn, Matplotlib
Estimated Cost	\$0
Deliverables	Slides, README, starter notebook, proposal PDF

This proposal is intentionally scoped for an “easy 90”: public dataset, standard model, free tools, and a realistic timeline.